

4.7 Solution: (a) Notice that

$$\sin^2 \theta = \frac{2}{3} (P_0(\cos \theta) - P_2(\cos \theta)),$$

we can write the charge density as

$$\rho(\mathbf{r}) = \frac{1}{96\pi} r^2 e^{-r} (P_0(\cos \theta) - P_2(\cos \theta)).$$

By Eq. (4.3), the multipole moments can be calculated as

$$\begin{aligned} q_{lm} &= \int Y_{lm}^*(\theta', \phi') r'^l \rho(\mathbf{x}') d^3 x' \\ &= \frac{1}{96\pi} \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} \int_0^{+\infty} r'^{l+4} e^{-r'} dr' \int (P_0(\cos \theta') - P_2(\cos \theta')) P_l^m(\cos \theta') e^{-im\phi'} d\Omega' \\ &= \frac{(l+4)!}{48} \sqrt{\frac{2l+1}{4\pi}} \int_{-1}^1 (P_0(x) - P_2(x)) P_l(x) dx, \end{aligned}$$

where we will have only $m = 0$ multipoles. The integral with respect to the Legendre polynomial is nonzero only for $l = 0$ and $l = 2$, and the nonvanishing multipole moments are

$$q_{00} = \frac{24}{48} \frac{1}{\sqrt{4\pi}} \times 2 = \frac{1}{\sqrt{4\pi}},$$

and

$$q_{20} = -\frac{720}{48} \sqrt{\frac{5}{4\pi}} \times \frac{2}{5} = 6\sqrt{\frac{5}{4\pi}}.$$

Applying Eq. (4.1), at large distances, the potential becomes, dropping sum over m ,

$$\begin{aligned} \Phi(\mathbf{x}) &= \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sqrt{\frac{4\pi}{2l+1}} q_{l0} \frac{P_l(\cos \theta)}{r^{l+1}} \\ &= \frac{1}{4\pi\epsilon_0} \left[\sqrt{4\pi} \cdot \frac{1}{\sqrt{4\pi}} \frac{1}{r} - \sqrt{\frac{4\pi}{5}} \cdot 6\sqrt{\frac{5}{4\pi}} \frac{P_2(\cos \theta)}{r^3} \right] \\ &= \frac{1}{4\pi\epsilon_0 r} \left(1 - \frac{6}{r^2} P_2(\cos \theta) \right). \end{aligned}$$

(b) The potential at an arbitrary point in space is given by

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sqrt{\frac{4\pi}{2l+1}} P_l(\cos \theta) \left[\int \frac{r'^l}{r^{l+1}} \rho(\mathbf{x}') Y_{lm}(\theta', \phi') d^3 x' \right].$$

To perform the integral, we need to break the integration range in r into two separate ranges. This can get quite tedious. Instead, we will directly calculate the potential for small r by setting r to 0 in the integration lower bound. After this, we will have

$$\Phi(\mathbf{x}) = \frac{1}{128\pi\epsilon_0} \sum_{l=0}^{\infty} r^l P_l(\cos \theta) \int_0^{+\infty} r'^{3-l} e^{-r'} dr' \cdot \frac{2}{3} \int_{-1}^1 (P_0(x) - P_2(x)) P_l(x) dx$$

$$\begin{aligned}
&= \frac{1}{128\pi\epsilon_0} \left[P_0(\cos\theta) \cdot 6 \cdot \frac{4}{3} - r^2 P_2(\cos\theta) \cdot \frac{4}{15} \right] \\
&= \frac{1}{4\pi\epsilon_0} \left[\frac{1}{4} - \frac{r^2}{120} P_2(\cos\theta) \right],
\end{aligned}$$

as expected.